

The Range of the Determinant Function on the Set of $n \times n$ (0, 1)-Matrices

R. Craigen
Department of Mathematics
University of Manitoba
Winnipeg, Manitoba
CANADA R3T 2N2
craigenr@cc.umanitoba.ca

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Abstract

As stated in [2], there is a conjecture that the determinant function maps the set of $n \times n$ (0, 1)-matrices onto a set of consecutive integers for any given n . While this is true for $n \leq 6$, it is shown to be false for $n = 7$. In particular there is no 7×7 determinant in the range $28 - 31$ but there is one equal to 32. Then the more general question of the range of the determinant function for all n is discussed. A lower bound is given on the largest string of consecutive integers centered at 0, each of which is a determinant of an $n \times n$ (0, 1)-matrix.

1 Introduction

Let us begin with a few conventions:

- Write Υ_n for the set of $n \times n$ (0, 1)-matrices.
- $\beta_n = \max_{M \in \Upsilon_n} |M|$.
- $D_k(n)$ is the assertion that there is a matrix in Υ_n whose determinant is k . Abusing the notation somewhat, any such matrix shall be denoted a $D_k(n)$.

Knowing the range of the determinant function on Υ_n may be useful, since the existence of a combinatorial design is equivalent to the existence of the corresponding (0, 1)-incidence matrix, and hence implies that the corresponding (usually predictable, as in the case of SBIBD's) determinant exists when that matrix is square.

Brenner and Cummings ([2], 1972) relate the following conjecture: that for a given n , the determinant function maps Υ_n onto a set of consecutive integers. In my notation, this translates to “ $|k| \leq \beta_n$ implies $D_k(n)$ ” for $n > 1$ (since we have $D_k(n)$ if and only if $D_{-k}(n)$ —see lemma 3—and consequently this set is symmetric with respect to 0). Subsequently we refer to this as the *Consecutive Integer Determinant Conjecture*.

Brenner says that he had raised the question with Marshall Hall in correspondence before 1972. He also relates that some time ago he heard someone had a counterexample. In any case I have not been able to find any that appear in print to date.

In section 2 we show that the above conjecture is in fact false. In section 3 we discuss some results on how large a set of consecutive integers k may be obtained satisfying $D_k(n)$. In the last section we outline some of the relevant unsolved questions, indicating some partial results obtained.

A final resolution to some of these questions may lead to a solution of the *Hadamard maximum determinant problem*, which is the question as to what is the maximum determinant among (± 1) -matrices of a given order ([2], [8])¹. This is because $D_k(n)$ is equivalent to the existence of a (± 1) -matrix of order $n + 1$ with determinant $2^n k$, modulo the following transformation:

$$\begin{aligned}
2^n |D| &= |2D| \\
&= \begin{vmatrix} 1 & 1 & \cdots & 1 \\ 0 & & & \\ \vdots & & 2D & \\ 0 & & & \end{vmatrix} \\
&= \begin{vmatrix} 1 & 1 & \cdots & 1 \\ -1 & & & \\ \vdots & & 2D - J & \\ -1 & & & \end{vmatrix},
\end{aligned} \tag{1}$$

where D is a $D_k(n)$ [11].²

Let us take note of a nice (well-known) property of rank-one matrices which will prove useful at several points. If U is rank-one, it is similar to an upper-triangular matrix. The only possible form for an upper-triangular rank-one matrix with trace equal to that of U is

$$\begin{pmatrix} \frac{\text{tr} U}{n} & * \\ 0 & \\ \vdots & 0 \\ 0 & \end{pmatrix}. \tag{2}$$

Noting that trace is similarity-invariant, immediately we have:

Lemma 1 (the “Rank-one Lemma”) *If U is rank-one, then $|I + U| = 1 + \text{tr}(U)$.*

2 The consecutive integer determinant conjecture is false

We shall see in the next section that the consecutive integer determinant conjecture holds for $n = 1, \dots, 6$. However, that is as far as it goes, as the following result shows:

Theorem 2 *There is no $D_k(7)$ for $27 < k < 32$.*

Proof: First note that $\beta_6 = 9$ (see section 3). This tells us that any matrix whose minors along a row are in Υ_6 , and whose entries in this row are from $\{0, \pm 1\}$, has determinant less than or equal

¹the *Hadamard bound*, $n^{\frac{n}{2}}$, holds in all orders. A (± 1) -matrix achieving this bound is called a *Hadamard matrix*. Equivalently, a Hadamard matrix H is one which satisfies $HH^t = nI$.

²and using the fact that we can negate rows and columns of a matrix without changing the absolute value of its determinant.

to 9 times the number of non-zero entries in this row. We shall use this observation repeatedly in the proof.

Now suppose A is a $D_k(7)$ with $k > 27$. Write $\mathbf{r}_1, \dots, \mathbf{r}_7$ for the rows of A , r_i for $\|\mathbf{r}_i\|$, and $\lambda_{i,j}$ for $\langle \mathbf{r}_i, \mathbf{r}_j \rangle$, $i \neq j$. Then proceed as follows:³

1. For each i , $r_i \geq 4$. This follows from the above observation.
2. Suppose $r_1 = 7$. Then:

$$A = \begin{matrix} \mathbf{r}_1 \\ \mathbf{r}_2 \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & * & * & * \\ & & & \vdots & & & \end{pmatrix} \sim \begin{pmatrix} 0 & 0 & 0 & 0 & * & * & * \\ & & & \vdots & & & \end{pmatrix}. \quad (3)$$

Here $\sim$ denotes equivalence via the action of adding a multiple of one row to another⁴. Plainly we have forced $|A| \leq 27$, which is a contradiction.

We argue similarly for each row, concluding that for each i , $r_i < 7$.

3. Suppose that $r_2 = r_3 = \dots = r_7 = 4$. Then for $i, j > 1$ we cannot have $\lambda_{i,j} = 3$ or 4 —else $|A| \leq 18$ as in 3. Thus $\lambda_{i,j} = 1$ or 2.

Now suppose $\lambda_{2,3} = 1$. Then:

$$\begin{matrix} \mathbf{r}_2 \\ \mathbf{r}_3 \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} \quad (4)$$

and we cannot have $\lambda_{2,j} = 1$ for $j > 3$, else then $\lambda_{3,j} = 3$ or 4, which, as we have just seen, cannot happen. Similarly, $\lambda_{3,j} \neq 1$, and so we have:

$$\begin{matrix} \mathbf{r}_2 \\ \mathbf{r}_3 \\ \mathbf{r}_j \end{matrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{pmatrix}. \quad (5)$$

Clearly, the first column of A must then have at most three nonzero entries. Expanding by minors along this column, we see that $|A| \leq 27$, which is a contradiction.

We conclude that if all $r_i = 4$ with at most one exception, then $\lambda_{i,j} = 2$ when $r_i = r_j = 4$.

4. Suppose that $r_1 = 6$. If in addition r_2 has a 0 in the same column as the 0 in $\mathbf{r}_1$, then:

$$A = \begin{matrix} \mathbf{r}_1 \\ \mathbf{r}_2 \end{matrix} \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & * & * \\ & & & \vdots & & & \end{pmatrix} \sim \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & * & * \\ & & & \vdots & & & \end{pmatrix} \quad (6)$$

This gives a contradiction as before. Thus, $\mathbf{r}_j$ must have a 1 in this column for $j > 1$.

It follows in the same manner that $r_j = 4$ for $j > 1$. Now by step 3, $\lambda_{i,j} = 2$ for $i, j > 1$.

Letting B be the submatrix obtained by excluding the first row and column of A , then, this all

³With no loss of generality, we assume at each step that the columns of A are arranged conveniently for display purposes.

⁴a determinant-preserving operation.

translates to $BJ = B^t J = 3J$ and $BB^t = 2I + J$. Thus $BB^t J = 9J = 2IJ + J^2 = 8J$, which is a contradiction (thus showing the well-known fact that an SBIBD(6,3,1) cannot exist).

We conclude that for each i , $r_i < 6$.

5. If $r_i = r_j = 5$, then $\lambda_{i,j} = 3$ —for if $\lambda_{i,j} = 5$ or 4 then $|A| = 0$ or ≤ 18 respectively, as in (3) and (6). Similarly, if $r_i = 5$ and $r_j = 4$ we cannot then have $\lambda_{i,j} = 3$ or 4 —else as above, $|A| \leq 27$ or 9 respectively. In any case, if $r_i = 5$ then every other row must have 1's in the columns corresponding to the 0's in $\mathbf{r}_i$. But then $|A| = 0$ —a contradiction.

We conclude that $r_i = 4$, all i .

6. Together, steps 3 and 5 tell us that $AA^t = 2I + 2J$.

We now can derive the following using the Rank-one Lemma:

$$\begin{aligned} |A| &= |AA^t|^{1/2} \\ &= |2I + 2J|^{1/2} \\ &= 2^{7/2} 8^{1/2} = 32 \end{aligned} \tag{7}$$

We have shown that if $A \in \Upsilon_7$ then either $|A| \leq 27$ or $|A| = 32$. The result follows. $\square$

The existence of a $D_{32}(7)$ may be inferred from the existence of a Hadamard matrix of order 8. This provides the claimed counterexample.

3 Proliferation of $(0, 1)$ -determinants

Here are some of the best bounds on β_n :

$$n \equiv 0 \pmod{4}: \beta_n \leq (2n+1)^{1/2} (n/4)^{n/2} \text{ (Barba, [1].)}$$

$$n \equiv 1 \pmod{4}: \beta_n \leq n((n-1)/4)^{\frac{n-1}{2}} \text{ (Ehlich [6], and Wojtas, [12].)}$$

Equality holds for $n \leq 100$ except $n = 21, 33, 57, 69, 73, 77, 89, 93, 97$ ([13] [3]).

$$n \equiv 2 \pmod{4}: \beta_n \leq (2n+1)^{1/2} (n/4)^{n/2} \text{ (Barba, [1].)}$$

$$n \equiv 3 \pmod{4}: \beta_n \leq 2^{-n} (n+1)^{\frac{n+1}{2}} \text{ (this follows from the Hadamard bound and (1)).}$$

This bound is achieved precisely when there exists a Hadamard matrix of order $n+1$ (this is established for many orders $\equiv 0 \pmod{4}$ including all less than 428 [9]).

- There is also the lower bound $\beta_n > 2^{-n} (3(n+1)/4)^{\frac{n+1}{2}}$ for all n (Clements and Lindstrom, [4]; Schmidt [10] gives a sharper, but less straightforward, lower bound).
- We have $\beta_n = 1, 1, 2, 3, 5, 9, 32, 56, 144, 320, 1458, 3645, 9531$ for $n = 1, 2, \dots, 13$ ([2], [11], [7]). $n = 14$ is apparently the first case for which β_n is not known.

The following main result for this section is given as one “monster lemma” since the different parts are designed to be used together recursively.

Lemma 3 (Monster Determinant Lemma, or MDL)

1. $D_k(n)$ iff $D_{-k}(n)$ for all $n > 1$
2. For any positive integers a, k and n , $D_k(n)$ implies $D_{ak}(n + a)$
3. If $D_k(n)$ and $D_l(m)$ then
 - (a) $D_{kl}(m + n)$
 - (b) $D_{k^m l^n}(mn)$
 - (c) $D_{2^{mn} l^{n+1} k^{m+1}}(mn + m + n)$
 - (d) if the former is r -row-regular⁵ and the latter has no row sum greater than r (for example, if it is s -row-regular with $s \leq r$ or if $m \leq r + 1$), then there is an r -row-regular $D_{kl}(m + n)$.
4. If there is an r -row-regular $D_k(n)$ and an s -row-regular $D_l(m)$ then
 - (a) there is an rs -row-regular $D_{k^m l^n}(mn)$
 - (b) there is a $(rm + sn - 2rs)$ -row-regular $D_{2^{(m-1)(n-1)} k^m l^n (\frac{n}{r}-2)^{m-1} (\frac{m}{s}-2)^{n-1} (\frac{n}{r} + \frac{m}{s} - 2)}(mn)$
 - (c) for $0 \leq d \leq n$, $0 \leq c \leq m$ there is a $D_{\frac{kl}{rs}(rs-cd)}(m+n)$ and a $D_{\frac{kl}{rs}(n(c-s)+m(d-r)+2rs-2cd)}(m+n-1)$
 - (d) for $-\min(r, s) \leq t \leq \min(n-r, m-s)$, there is an $(r+s+t)$ -row-regular $D_{\frac{kl}{rs}t(r+s+t)}(m+n)$.
5. If there is an r_i -row-regular $D_{k_i}(n_i)$ for each $i = 1, \dots, s$ then
 - (a) $D_{k_1 \dots k_s (\frac{a_1}{r_1} + \dots + \frac{a_s}{r_s} - 1)}(n_1 + \dots + n_s)$ for a_i satisfying $0 \leq a_i \leq n_i$
 - (b) $D_{k_1 \dots k_s (\frac{n_1}{r_1} + \dots + \frac{n_s}{r_s} - 2)}(n_1 + \dots + n_s - 1)$.
6. There is a 1-regular $D_1(1)$.
7. If there is an r -row-regular $D_k(n)$ then there is an $(n - r)$ -row-regular $D_{k(\frac{n}{r}-1)}(n)$
8. If there is a Hadamard matrix of order $n + 1$ (for example, any $n \equiv 3 \pmod{4}$ and ≤ 407) then there is an $(n + 1)/2$ -regular $D_{2^{(\frac{n+1}{4})} \frac{n+1}{2}}(n)$
9. If there is a Hadamard matrix of order n and excess⁶ x then $D_{(\frac{n}{4})^{\frac{n}{2}} (1+x/n)}(n)$.

Proof:

⁵a matrix R is r -row(respectively column)-regular if $RJ = rJ$ (respectively $JR = rJ$). A matrix is r -regular if it is both r -row and r -column-regular. In the lemma we shall disallow 0-row-regularity whenever this would imply division by 0.

⁶The excess of a matrix is the sum of its entries. Many such cases may be inferred from results in [5] —for example, all $x \equiv 0 \pmod{4}$ satisfying $|x| \leq \max(n, 3n - 8)$, and all $x = 2a\sqrt{n}$ for $|a| \leq n/2$ when n is the order of a regular Hadamard matrix, as is the case for each n where $\sqrt{n}$ is the order of a Hadamard matrix.

1. This follows from the fact that interchanging two rows or columns negate a determinant (henceforth, this fact will be used without reference).
2. Suppose A is a $D_k(n)$. Expanding the following array by minors will verify that it is a $D_{\pm ak}(n+a)$:

$$\left(\begin{array}{c|ccc|cccc} 0 & 1 & \cdots & 1 & 1 & 0 & \cdots & 0 \\ \hline 1 & & & & & & & \\ \vdots & & & & & & & \\ 1 & & & & & & & 0 \\ \hline \mathbf{v} & & & 0 & & & & A \end{array} \right) \quad (8)$$

Here $\mathbf{v}$ is the first column of A .

3. Let A be a $D_k(n)$ and B be a $D_l(m)$. Then

$$(a) \quad |A \oplus B| = \begin{vmatrix} A & 0 \\ 0 & B \end{vmatrix} = |A||B|.$$

$$(b) \quad |A \otimes B| = |A|^m |B|^n.$$

- (c) using (1), Construct (± 1) -matrices A' of order $n+1$ and B' of order $m+1$. Then verify that $A' \otimes B'$ is a (± 1) -matrix of order $mn+m+n+1$ with determinant $(2^n k)^{m+1} (2^m l)^{n+1}$. Then (1) implies the existence of the required matrix in Υ_{mn+m+n} .
- (d) If the row sums of B are all less than or equal to r , then we can choose C in Υ_n such that $B+C$ is r -row-regular, so then $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix}$ an r -row-regular $D_{kl}(m+n)$.

4. Now further suppose that A is r -row-regular and B is s -row-regular. then

- (a) consider $A \otimes B$.

- (b) apply the Rank-one lemma three times to $\frac{J_{mn} - (2A - J_n) \otimes (2B - J_m)}{2}$.

- (c) choose C to be c -row-regular and D to be rank one and d -row-regular. Let $X = \begin{pmatrix} A & C \\ D & B \end{pmatrix}$. Using the Rank-one lemma (with $X = Y + \begin{pmatrix} 0 & 0 \\ D & 0 \end{pmatrix}$), we get $|X| = \frac{kl}{rs}(rs - cd)$, which is the first determinant. The reader may verify that $X^{-1} = Y^{-1} + \frac{rs}{cd-rs} Y^{-1} \begin{pmatrix} 0 & 0 \\ D & 0 \end{pmatrix} Y^{-1}$. Thus, $tr(X^{-1} J_{m+n}) = \frac{n(c-b)+m(d-a)}{cd-rs}$. From this we may use the Rank-one lemma again to show that $2X - J_{m+n}$ is a (± 1) -matrix with determinant $2^{m+n} \frac{kl}{rs} \left(\frac{n(c-s)+m(d-r)}{2} + rs - cd \right)$, which together with (1) establishes the second determinant.

- (d) Now if $-\min(r, s) \leq t \leq \min(n-r, m-s)$, we can find c and d as above with $r+c = s+d = r+s+t$. Then X is $r+s+t$ -row-regular and has determinant $-\frac{kl}{rs} t(r+s+t)$.

5. Now let $X_1, \dots, X_s$ be, respectively, r_i -row-regular $D_{k_i}(n_i)$'s for $i = 1, \dots, s$. For compactness, write $J_{m,n}^a$ for the $m \times n$ matrix whose entries in the first a columns are all 1 and the rest of whose entries are 0. Then take

$$M = \begin{pmatrix} X_1 & & 0 \\ & \ddots & \\ 0 & & X_s \end{pmatrix} \text{ and } U = \begin{pmatrix} J_{n_1 \times n_1}^{a_1} & \cdots & J_{n_1 \times n_s}^{a_s} \\ \vdots & & \vdots \\ J_{n_s \times n_1}^{a_1} & \cdots & J_{n_s \times n_s}^{a_s} \end{pmatrix}. \quad (9)$$

(a) It is easy to verify that

$$M^{-1}U = \begin{pmatrix} \frac{1}{r_1} J_{n_1 \times n_1}^{a_1} & \cdots & \frac{1}{r_1} J_{n_1 \times n_s}^{a_s} \\ \vdots & & \vdots \\ \frac{1}{r_s} J_{n_s \times n_1}^{a_1} & \cdots & \frac{1}{r_s} J_{n_s \times n_s}^{a_s} \end{pmatrix}. \quad (10)$$

Thus $\text{tr}(M^{-1}U) = \frac{a_1}{r_1} + \cdots + \frac{a_s}{r_s}$, and so $|M - U| = |M||I - M^{-1}U| = k_1 \cdots k_s (1 - (\frac{a_1}{r_1} + \cdots + \frac{a_s}{r_s}))$, by an application of the Rank-one lemma. Negating the appropriate columns gives a matrix in Υ_n whose determinant has the same norm.

(b) When $a_i = n_i$, all i , then $U = J$, so $\text{tr}(M^{-1}J) = \frac{n_1}{r_1} + \cdots + \frac{n_s}{r_s}$. Apply (1) to $2M - J$.

6. This is clear.

7. Now let A be an r -row-regular $D_k(n)$. Then $J - A$ is $(n - r)$ -row-regular and the Rank-one lemma gives $|J - A| = \pm |A||I - A^{-1}J| = \pm k(1 - \text{tr}(\frac{1}{r}J)) = \pm k(\frac{n}{r} - 1)$.

8. this follows directly from the construction in (1), and the fact that the given Hadamard matrix has determinant $(n + 1)^{\frac{n+1}{2}}$.

9. Apply the Rank-one lemma to $\frac{J+H}{2}$, where H is the given Hadamard matrix.

□

The following two theorems are just a simple demonstration of the power of this lemma:

Theorem 4 *For each n and $0 \leq k < n$, there is a k -row-regular $D_k(n)$.*

Proof: I_{k+1} is a 1-regular $D_1(k+1)$ a k -regular $D_k(k+1)$ follows by part 7 of MDL. The result follows by using this matrix together with I_{n-k-1} in part 3d). □

Theorem 5 *If there is an r -row-regular $D_k(n)$ then $r|k$.*

Proof: Using this matrix and I_1 in part 3d), we get an r -row-regular $d_k(n+1)$. Thus there is a $D_{k(\frac{n}{r}-1)}(n)$ and a $D_{k(\frac{n+1}{r}-1)}(n+1)$, by part 7. So $r|kn$ and $r|k(n+1)$ and so $r|k$. □

Table 1 compiles the results given by MDL for $n \leq 10$, comparing them to determinants found using other methods (“–” indicates an entire range of determinants which are established, the left endpoint defaulting to 0).

Table 1: Determinants found up to order 10 with and without MDL

n	MDL implies $D_k(n)$ for $k =$	other determinants found without MDL
1	-1	
2	-1	
3	-2	
4	-3	
5	-5	
6	-8	9
7	-13, 16, 18, 20, 24, 32	14, 15, 17, 19
8	-18, 20, 24, 28, 32, 40, 48, 56	19, 21 - 23, 25 - 27, 29 - 31, 33 - 39, 42, 44, 45
9	-24, 26, 28, 32, 40, 48, 56, 64, 72, 80, 88, 144	25, 27, 29 - 31, 33 - 39, 41 - 47, 49 - 55, 57 - 63, 65 - 71, 73 - 79, 89 - 102, 104, 105, 108, 110, 112, 116, 120, 125, 128
10	-34, 36, 38 - 40, 44, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, 136, 144	35, 37, 41 - 43, 45 - 47, 49 - 55, 57 - 63, 65 - 71, 73 - 79, 81 - 87, 89 - 95, 97 - 103, 105 - 111, 113 - 119, 121 - 127, 129 - 135, 137 - 143, 145 - 232, 234 - 256, 258, 260, 261, 263 - 267, 270, 272 - 276, 279, 280, 283 - 285, 288, 291, 294 - 297, 304, 312, 315, 320

4 What next?

With the demise of the consecutive integer determinant conjecture, several related questions are left wide open:

- What is the range of the determinant function on Υ_n in general, and the value of β_n in particular?
- It appears from constructions so far that very strong structure is dictated for matrices of large determinant in Υ_n . For example, if a Hadamard matrix of order $n + 1$ exists, a $D_{\beta_n}(n)$ must be an SBIBD($n, \frac{n+1}{2}, \frac{n+1}{4}$) (or rather, the $(0, 1)$ -incidence matrix corresponding to it—however, we shall not distinguish between them here). Now consider a matrix from Υ_n of the form $\left(\begin{array}{ccc|ccc} 1 & \cdots & 1 & 0 & \cdots & 0 \\ \hline & A & & & B & \end{array} \right)$. We can show that it has, up to sign, the same determinant as $\left(\begin{array}{ccc|ccc} 1 & \cdots & 1 & 0 & \cdots & 0 \\ \hline & J - A & & & B & \end{array} \right)$. If we combine transformations of this type with row and column permutation and the transpose operation, these together define an equivalence relation in Υ_n with respect to which the absolute value of the determinant is invariant. With this notion of equivalence it is possible to show that $D_1(2)$ is equivalent to I_2 , a $D_2(3)$ is equivalent to $J_3 - I_3$ (that is, an SBIBD(3, 2, 1)), a $D_3(4)$ is equivalent to an SBIBD(4, 3, 2),

or alternately, $\left(\begin{array}{c|ccc} 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array} \right)$. Similarly, a $D_5(5)$ is equivalent to $\left(\begin{array}{cc|ccc} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array} \right)$.

Are there principles which dictate, up to equivalence, the structure of a $D_{\beta_n}(n)$, or in general any “large enough” determinant in any given order?

- With what frequency does each determinant occur in each order? Figure 1 compiles results obtained from a fairly large sample in order 7.

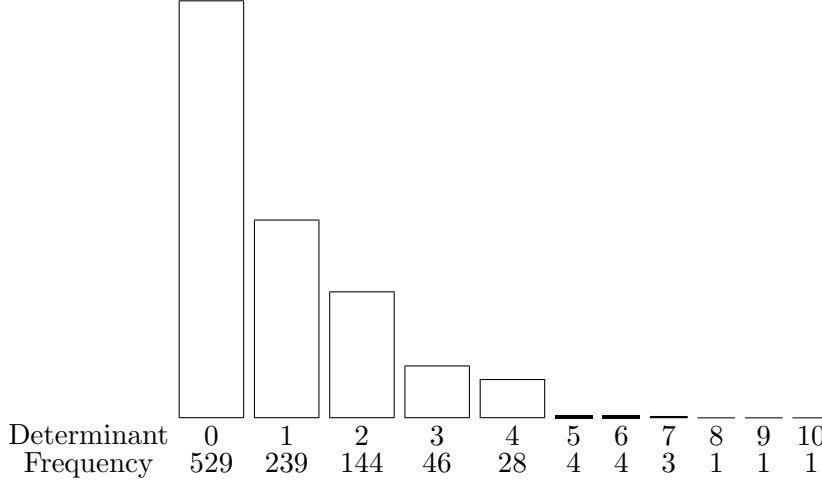


Figure 1: Occurrences of determinants of order 7 among a sample of 1000

- With regard to the relationship between row-regularity and determinants, there are a number of questions. Chief among these is the following: for what triples (r, n, k) do there exist an r -row-regular $D_k(n)$? Some elementary partial results:
 - (r, n, k) must satisfy $0 \leq r \leq n$, $k \leq \beta_n$, $r|k$. If $r = 0$ or n , $k = 0$; if $r = 1$, $k = 0$ or 1 ; if $r = n-1$, $k = 0$ or $n-1$.
 - if $r|k$ and $D_p(r+1)$ for every factor p in some factorization of k then there is an n for which (r, n, k) is such a triple (this is almost certainly true when the second condition is replaced with $r > 1$). In particular, when $r \geq \sqrt{k}$ we may take $n \leq 2k$.
 - if (r, n, k) is such a triple and $n' > n$ then (r, n', k) is one as well.
- For each n , let β'_n be the largest number such that $k \leq \beta'_n$ implies $D_k(n)$. In other words, $[-\beta'_n, \beta'_n]$ is the largest possible interval of consecutive integer $(0, 1)$ -determinants in order n centered at 0. We have shown that $\beta'_7 \neq \beta_7$. Can we have $\beta'_n = \beta_n$ for $n > 7$? What are some non-trivial bounds on β'_n ? The only *upper* bound I know so far is β_n . A trivial lower bound from theorem 4 is $\beta'_n \geq n-1$. Here is a somewhat better one:

Theorem 6 For all n , $\beta'_n \geq \lfloor \frac{n}{2} \rfloor - 1$.

Proof: Write $u = \lfloor \frac{n}{2} \rfloor$. If $m \leq u$ then theorem 4 provides a $D_m(n)$. If $u < m \leq u^2 - 1$, we can write $m = s(u-1) + t$, where $1 \leq s \leq u$ and $1 \leq t \leq u$. Now take an s -row-regular $D_s(u)$ and I_{n-u} and apply part 5a) of MDL, with $a_1 = t$ and $a_2 = u$. This gives us a $D_{1 \cdot s(\frac{t}{s} + \frac{u}{1} - 1)}(u + (n-u)) = D_m(n)$. $\square$

This theorem actually shows that part 2 of MDL is not necessary for $a > 4$, since then $\beta'_a > a$, and so the first point of part 3 gives the same result (we note that part 2 is not used to establish this result).

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